DOCTORAL RESEARCH PROPOSAL

Topological Asteroseismology

Persistent structure in light curves, power spectra, and échelle geometry of solar-like stars

Developing stable and physically interpretable topological observables for solar-like oscillations, with KIC 10963065 ("Rudy") as the benchmark and the Kepler LEGACY sample as the population-level validation set.

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Version Research planning edition – July 2026

Proposal status and intended use

This document is a detailed research-design proposal. It defines the scientific hypotheses, mathematical constructions, validation logic, work packages, stage gates, expected publications, and reproducibility requirements for a doctoral project. Numerical values listed for the benchmark star are literature or pipeline reference values and must be independently reproduced before being treated as project results. The final institutional version should replace the bracketed affiliation fields and be adapted to the rules of the relevant graduate program and funding agency.

Recommended citation for internal use: Wendell, G. (2026), *Topological Asteroseismology of Solar-Like Stars: Persistent Structure in Light Curves, Power Spectra, and Échelle Geometry*, doctoral research proposal.

Abstract

Solar-like oscillations provide a direct probe of stellar interiors and allow precise inference of global stellar properties, internal stratification, age-sensitive frequency ratios, rotation, and acoustic glitches. Conventional analyses exploit the frequency of maximum oscillation power $\nu_{\max}$, the large separation $\Delta\nu$, individual mode frequencies, and the ridge morphology of échelle diagrams. These methods are mature, but their performance degrades when observation duration, duty cycle, signal-to-noise ratio, or spectral resolution is limited. They also summarize oscillation spectra primarily through local peak measurements, autocorrelations, or model-dependent fits. Topological data analysis (TDA), particularly persistent homology, offers a complementary multiscale description of shape and connectivity with formal stability guarantees. Its use in solar-like asteroseismology, however, remains essentially unexplored.

This project will develop a physically interpretable framework for topological asteroseismology. The framework will operate on five related representations: one-dimensional light-curve filtrations, time-delay embeddings, background-normalized power spectra, mode-frequency point clouds, and a new cylindrical formulation of the échelle diagram. The principal methodological contribution will be a *topological échelle diagram*: an intensity field and ridge graph on $S^1 \times I$ whose persistent connected structures quantify mode-ridge coherence as a function of trial $\Delta\nu$. The project will investigate topological estimators of $\nu_{\max}$ and $\Delta\nu$, angular representations of small frequency-separation ratios, persistence-based measures of mode coherence, and analytical bounds linking photometric or spectral perturbations to persistence diagrams and, through seismic scaling relations, to stellar-parameter uncertainty.

KIC 10963065 (Rudy; HD 176693; Kepler-408; TIC 48450369) will serve as the controlled benchmark because it has Kepler short-cadence observations, published individual mode frequencies and stellar properties, reproducible Lightkurve baselines, and multiple TESS sectors. Method development will not rely on this star alone. Ground-truth validation will combine stochastic oscillator simulations, MESA/GYRE-based mode spectra, the 66-star Kepler LEGACY sample, matched control stars, and Kepler-to-TESS degradation experiments. Topological results will be evaluated against Lightkurve, pySYD, autocorrelation methods, conventional échelle analysis, and published peak-bagging values. Phase-randomized and IAAFT surrogates will test whether time-domain topology contains information beyond the power spectrum. Hybrid classical-topological models will be accepted only when they improve out-of-sample parameter recovery, uncertainty calibration, or reliability assessment after conditioning on conventional metrics.

The expected outputs are two or three astronomy papers, an optional mathematical-methods paper, a reproducible open-source pipeline, benchmark datasets, and a validated decision framework identifying when TDA adds genuine information and when it merely re-encodes conventional spectral structure. The core scientific claim is therefore deliberately falsifiable: persistent topology should either improve or certify asteroseismic inference under realistic observing limitations, or the project will quantify the conditions under which it does not.

Executive summary

Central thesis

Persistent structures associated with oscillation peaks, folded mode ridges, and delay-reconstructed dynamics can provide stable and physically interpretable observables of solar-like oscillations. When appropriately validated, these observables can complement conventional asteroseismic parameters, improve their reliability under degraded observing conditions, and connect formal stability theorems from TDA to uncertainty in stellar properties.

Core innovation. The proposal does not treat TDA as a generic feature generator. It defines a physics-first construction on the natural cylindrical domain of an échelle diagram. For a trial separation Δ, the background-normalized spectrum is folded into an intensity field $E_\Delta(q, \theta)$ on radial order q and circular phase θ. Persistent ridge connectivity, cubical persistence, and zigzag persistence across order will be used to score ridge coherence. The optimal score defines a candidate topological estimator $\widehat{\Delta\nu}^{\mathrm{top}}$. Angular ridge separation provides a natural geometric link to r_{02} and related frequency ratios.

Validation principle. Rudy is the benchmark, not the population. The project moves from controlled simulations to a single-star case study, then to the Kepler LEGACY sample and TESS transfer tests. Every proposed topological statistic will be evaluated against physically meaningful nulls and strong classical baselines. A result is considered scientifically successful if it achieves at least one of three outcomes:

S1. improves out-of-sample recovery of an asteroseismic or stellar parameter;

S2. provides calibrated uncertainty or a reliable quality-control diagnostic under adverse observing conditions;

S3. yields a new analytical relation or stability bound with explicit physical interpretation.

Planned publication units. The intended publication architecture is: (I) persistent topology of solar-like oscillation spectra and benchmark validation; (II) topological échelle geometry, small separations, and mathematical stability; and (III) hybrid Kepler–TESS inference and reliability. The project is designed so that each paper has its own falsifiable question and independent deliverables.

Primary risk. For a linear stationary Gaussian process, the power spectrum determines the second-order stochastic structure; delay-embedding topology may therefore add little beyond spectral information. This is treated as a scientific hypothesis to test, not an inconvenience to hide. Phase-randomized surrogates, ablations, and conditional-information analyses are built into the core design.

Contents

Abstract iii

Executive summary iv

Abbreviations and notation x

1. Motivation, context, and significance 1
- 1.1. Scientific motivation . . . 1
- 1.2. Why the problem is timely . . . 1
- 1.3. Benchmark system: KIC 10963065 . . . 2
- 1.4. Astronomical and methodological significance . . . 2

2. Literature review and research gap 4
- 2.1. Solar-like oscillations and global asteroseismology . . . 4
- 2.2. Pipelines, spectra, and échelle diagrams . . . 4
- 2.3. Persistent homology and stable topological summaries . . . 5
- 2.4. Topology of time series and periodic signals . . . 5
- 2.5. TDA in astronomy and adjacent work . . . 5
- 2.6. Research gap . . . 6

3. Research questions, hypotheses, objectives, and scope 7
- 3.1. Primary research question . . . 7
- 3.2. Specific research questions . . . 7
- 3.3. Hypotheses and falsification criteria . . . 7
- 3.4. General objective . . . 8
- 3.5. Specific objectives . . . 8
- 3.6. Scope and exclusions . . . 8
- 3.7. Project architecture . . . 9

4. Mathematical and physical framework 10
- 4.1. Solar-like oscillations as a physical inference problem . . . 10
  - 4.1.1. Linear adiabatic oscillations . . . 10
  - 4.1.2. Asymptotic p-mode pattern . . . 10
  - 4.1.3. Stochastic excitation, damping, and background . . . 11
  - 4.1.4. Scaling relations and propagated uncertainty . . . 11
- 4.2. Persistent homology and stable summaries . . . 11
  - 4.2.1. Filtered spaces and persistence diagrams . . . 11
  - 4.2.2. Stability as an astrophysical quality guarantee . . . 12
- 4.3. Five complementary topological constructions . . . 12
  - 4.3.1. A. Sublevel and superlevel topology of the light curve . . . 12
  - 4.3.2. B. Sliding-window or delay-coordinate geometry . . . 12
  - 4.3.3. C. Superlevel topology of the background-normalized PSD . . . 13
  - 4.3.4. D. Topology and geometry of extracted mode frequencies . . . 13
  - 4.3.5. E. The topological échelle diagram . . . 13
- 4.4. Analytical relationships and theorem targets . . . 14
  - 4.4.1. A concrete lower bound on ridge persistence . . . 15

4.5. Interpretive limits . . . 15

5. Data, experiments, and computational methodology 16
5.1. Study design . . . 16
5.2. Benchmark star: KIC 10963065 (Rudy) . . . 16
5.3. Acquisition, provenance, and data quality . . . 17
5.4. Preprocessing and sensitivity branches . . . 17
5.5. Classical asteroseismic baseline . . . 18
5.6. Simulation program . . . 18
5.6.1. Fast stochastic simulator . . . 18
5.6.2. Experimental factors . . . 18
5.6.3. Physics-informed stellar models . . . 19
5.7. Topology pipeline . . . 19
5.7.1. Representation-specific computation . . . 19
5.7.2. Algorithm for the topological échelle estimator . . . 19
5.7.3. Parameter selection without leakage . . . 20
5.8. Surrogate tests and null hypotheses . . . 20
5.9. Inference, uncertainty, and comparison metrics . . . 20
5.9.1. Uncertainty propagation . . . 20
5.9.2. Performance metrics . . . 21
5.9.3. Hybrid models . . . 21
5.10. Validation protocol and power . . . 21
5.11. Ablations, robustness, and negative controls . . . 22
5.12. Software architecture and reproducibility . . . 22
5.13. Computational resources . . . 22

6. Work packages, deliverables, and decision gates 24
6.1. Work-package logic . . . 24
6.1.1. WP0 – Governance, literature map, and reproducibility foundation . . . 24
6.1.2. WP1 – Classical benchmark and observation model . . . 24
6.1.3. WP2 – Analytic cases and stochastic simulation laboratory . . . 24
6.1.4. WP3 – Representation comparison and estimator development . . . 24
6.1.5. WP4 – Rudy end-to-end topological case study . . . 25
6.1.6. WP5 – Population validation on Kepler LEGACY stars . . . 25
6.1.7. WP6 – Theory and topological échelle formalization . . . 25
6.1.8. WP7 – Hybrid inference and TESS transfer . . . 25
6.1.9. WP8 – Synthesis, release, and thesis . . . 26
6.2. Stage gates . . . 26

7. Expected results and publication strategy 27
7.1. Expected scientific results . . . 27
7.2. Quantitative success criteria . . . 27
7.3. Anticipated figures and tables . . . 28
7.4. Publication plan . . . 29
7.5. Value of negative results . . . 29

8. Risks, limitations, and research integrity 30
8.1. Risk register . . . 30
8.2. Known scientific limitations . . . 31
8.3. Research integrity and open-science plan . . . 31
8.4. Data-management plan . . . 32

9. Schedule and agenda **33**
9.1. Calendar alignment and doctoral critical path 33
9.2. Full thirty-month publication program 34
9.3. Milestones and review agenda 36
9.4. Recurring working rhythm 37

10. Conclusion **38**

A. Notation and operational definitions **39**

B. Rudy benchmark checklist **40**

C. Minimum simulation and ablation matrix **41**

D. Reproducibility and reporting checklist **42**

E. Future extensions outside the critical path **43**

References **44**

List of Figures

3.1. Research architecture. Simulations and observations feed the same representation layer; topological quantities are compared with strong classical baselines before any physical claim is made. 9

4.1. Computational definition of the topological échelle diagram. Periodicity is respected by the cylindrical metric; persistent structures are converted to physical quantities only after null calibration. 14

9.1. Thirty-month Gantt plan. Continuous reproducibility work and overlapping writing reduce end-loaded risk. 35

List of Tables

4.1. Planned mathematical results and their astrophysical use. 14

5.1. Data tiers and inferential roles. Catalog holdings will be frozen in a versioned manifest at project start. 16
5.2. Primary topological representations and controls. 19

6.1. Decision gates that control scope and protect the critical path. 26

7.1. Primary success criteria. Final numerical margins are frozen after WP2 pilots and before validation. 28
7.2. Planned manuscripts. Titles are provisional. 29

8.1. Technical and programmatic risks. Probability (P) and impact (I) are rated low (L), medium (M), or high (H). 30

9.1. Accelerated doctoral critical path to a May 2027 defense. 33
9.2. Milestones, evidence, and review decisions. 36

A.1. Principal symbols and operational definitions. 39

C.1. Core experiments that cannot be omitted from the thesis. 41

Abbreviations and notation

Symbol / term	Meaning
TDA	Topological data analysis.
PH	Persistent homology.
PSD	Power spectral density.
$\nu_{\max}$	Frequency at which the oscillation power envelope is maximal.
$\Delta\nu$	Large frequency separation between modes of equal degree and adjacent radial order.
$\delta\nu_{02}$	Small separation between radial and quadrupole modes.
r_{02}, r_{01}, r_{10}	Frequency-separation ratios used in detailed stellar modelling.
D_k or Dgm_k	Persistence diagram in homological dimension k.
H_k	Homology group in dimension k.
d_{B}	Bottleneck distance between persistence diagrams.
VR	Vietoris–Rips filtration.
SW	Sliding-window or time-delay embedding.
KIC / TIC	Kepler Input Catalog / TESS Input Catalog.
LEGACY	Kepler Asteroseismic LEGACY sample of 66 well-characterized solar-like stars.
IAAFT	Iterative amplitude-adjusted Fourier-transform surrogate.
MESA / GYRE	Stellar evolution and stellar oscillation software used to produce physical simulations.
SPOC	Science Processing Operations Center pipeline for TESS products.
WP	Work package.

1. Motivation, context, and significance

1.1. Scientific motivation

Stars are the fundamental clocks, laboratories, and population tracers of astrophysics, yet many of their most important properties cannot be measured directly. Mass, radius, age, internal rotation, convective-boundary structure, helium abundance, and the sound-speed profile are inferred through combinations of spectroscopy, astrometry, stellar evolution models, and oscillation data. Asteroseismology is uniquely powerful because global stellar oscillations respond to conditions throughout the stellar interior. Solar-like oscillators – cool main-sequence stars, subgiants, and red giants with outer convective envelopes – display stochastically excited and intrinsically damped modes whose frequencies and line profiles encode global and local structure [Aerts et al., 2010, Chaplin and Miglio, 2013, García and Ballot, 2019].

The Kepler mission transformed this field by delivering long, high-duty-cycle photometric time series with sufficient precision to resolve oscillation spectra for thousands of stars and individual modes for a benchmark subset [Borucki et al., 2010, Lund et al., 2017]. TESS expanded the observational landscape to much of the sky, but its typical sector length, larger pixels, and discontinuous revisits make detailed solar-like asteroseismology more challenging for main-sequence targets [Ricker et al., 2015]. The coming era therefore creates two simultaneous needs: methods that exploit the information content of the best legacy data, and methods that remain reliable when duration, cadence, duty cycle, or signal-to-noise is less favorable.

Conventional global asteroseismology is based mainly on $\nu_{\max}$, $\Delta\nu$, and empirical or theoretically motivated scaling relations. Detailed modelling uses individual mode frequencies, frequency ratios, rotational splittings, and acoustic glitches. These methods are physically mature, but their extraction procedures may involve smoothing scales, background models, peak-detection thresholds, initial scaling-relation guesses, or computationally intensive peak bagging. An échelle diagram provides a powerful visual representation of the mode comb, yet the conventional diagram is generally interpreted visually or through ridge fitting rather than through a multiscale invariant.

TDA offers a different language. Instead of identifying a fixed collection of peaks at a single threshold, persistent homology records how connected components, loops, and higher-dimensional structures appear and disappear as a scale or intensity threshold changes [Edelsbrunner et al., 2002, Zomorodian and Carlsson, 2005, Carlsson, 2009]. Persistence diagrams possess formal stability guarantees, and time-delay embeddings permit oscillatory signals to be represented geometrically [Cohen-Steiner et al., 2007, Perea and Harer, 2015]. The scientific opportunity is not merely to "apply TDA" to stellar data, but to construct filtrations whose births, deaths, and connecting structures have direct asteroseismic meaning.

1.2. Why the problem is timely

The project is timely for five reasons.

1. **Rich ground truth exists.** The Kepler LEGACY sample provides published individual frequencies and precise stellar properties for 66 main-sequence solar-like stars [Lund et al., 2017, Silva Aguirre et al., 2017].

2. **Open baselines exist.** Lightkurve and pySYD make conventional global analysis reproducible and suitable for direct comparison [Lightkurve Collaboration et al., 2018, Chontos et al., 2022].

3. **Cross-mission data exist.** Several Kepler targets, including KIC 10963065, have TESS observations, enabling controlled transfer and degradation tests [Mikulski Archive for Space Telescopes, 2026].
4. **TDA is statistically mature enough to test.** Stable summaries, persistence landscapes, persistence images, efficient Rips implementations, and cubical-complex software are available [Bubenik, 2015, Adams et al., 2017, Bauer, 2021, Tauzin et al., 2021].
5. **The methodological gap is identifiable.** Previous astronomy applications demonstrate that persistence can complement conventional summaries, while existing TDA work on Kepler light curves has focused on transit classification rather than solar-like seismic observables [Heydenreich et al., 2021, Chung, 2025].

1.3. Benchmark system: KIC 10963065

KIC 10963065 is also known as Rudy, HD 176693, Kepler-408, and TIC 48450369. It is a bright F-type solar-like oscillator and planet-host star in the Kepler LEGACY sample. It is an unusually useful benchmark because the following independent resources are available:

- Kepler short-cadence light curves spanning multiple quarters;
- published peak-bagged oscillation frequencies and frequency ratios;
- detailed stellar-model inferences and recent acoustic-glitch analysis;
- a reproducible Lightkurve tutorial using quarters 2, 5, 6, and 7;
- TESS target light curves in multiple observing sectors;
- a close-orbiting planet whose transits can be explicitly masked when constructing stellar variability products.

The Lightkurve tutorial obtains reference values of approximately

$$\nu_{\max} \simeq 2145\,\mu\text{Hz}, \qquad \Delta\nu \simeq 103.11\,\mu\text{Hz}, \tag{1.1}$$

and uncorrected scaling-relation estimates of $M \simeq 1.08 M_\odot$, $R \simeq 1.23 R_\odot$, and $\log g \simeq 4.29$ [Hall and Barentsen, 2020]. These are reference targets for pipeline reproduction, not assumed truths. The project will compare them with published frequencies and with pySYD estimates.

> **Scope discipline.** Rudy is too well studied for a single-star descriptive analysis to be a strong primary publication. A 2026 study already compares acoustic-glitch indicators and stellar models for Rudy, Perky, and 16 Cyg A [Özel et al., 2026]. In this proposal Rudy is used as a controlled benchmark, visualization case, and cross-mission test. General scientific claims require simulations and population validation.

1.4. Astronomical and methodological significance

If successful, the project would contribute at three levels.

Observational astronomy.
It would provide robust diagnostics for detecting and characterizing solar-like oscillations under limited duration, irregular windows, or lower signal-to-noise.

Stellar physics.
It would connect persistent structures to mean density, acoustic travel time, small-separation ratios, mode coherence, and potentially glitch or rotational morphology.

Applied mathematics.
It would formulate persistence constructions on an échelle cylinder and derive stability or approximation bounds for noisy mode combs and damped oscillatory signals.

The negative outcome is also informative. If time-domain topology contains no conditional information once the PSD is known, the project will quantify that limitation and identify the representations where TDA is useful primarily for stability or quality control rather than additional physics.

2. Literature review and research gap

2.1. Solar-like oscillations and global asteroseismology

Solar-like oscillations are excited by turbulent convection and damped by near-surface processes. In the power spectrum, each resolved mode is approximately Lorentzian, while mode amplitudes are modulated by a broad power-excess envelope around $\nu_{\max}$. The background contains contributions from granulation, activity, instrumental systematics, and photon noise [Chaplin and Miglio, 2013, García and Ballot, 2019, Kallinger et al., 2014]. The approximately regular p-mode frequency pattern follows asymptotic theory [Tassoul, 1980] and motivates the use of large and small separations.

The large separation is related to the inverse acoustic diameter,

$$\Delta\nu \simeq \left(2\int_0^R \frac{dr}{c_s(r)}\right)^{-1}, \tag{2.1}$$

and scales approximately with the square root of mean density. The frequency $\nu_{\max}$ scales approximately with acoustic cutoff frequency and therefore with $g/\sqrt{T_{\rm eff}}$. Combining the two produces widely used mass, radius, density, and surface-gravity scaling relations [Kjeldsen and Bedding, 1995, Huber et al., 2011, Hekker, 2020]. These relations are powerful but not exact; corrections depend on evolutionary state, metallicity, temperature, and model physics.

Detailed inference proceeds beyond global parameters. Small separations and ratios such as r_{02} and r_{010} reduce sensitivity to poorly modelled surface layers and constrain the sound-speed gradient near the core. Rotationally split multiplets constrain internal rotation and inclination. Oscillatory signatures in frequencies or frequency differences diagnose acoustic glitches caused by helium ionization zones and convective boundaries. The LEGACY analyses provide a homogeneous foundation for these measurements [Davies et al., 2016, Lund et al., 2017, Silva Aguirre et al., 2017].

2.2. Pipelines, spectra, and échelle diagrams

Lightkurve provides accessible tools for Kepler and TESS time-series retrieval, correction, periodograms, global seismic estimation, and conventional échelle visualization [Lightkurve Collaboration et al., 2018, Hall and Barentsen, 2020]. pySYD reproduces the well-tested SYD methodology in an open Python implementation and estimates $\nu_{\max}$, $\Delta\nu$, background components, amplitudes, and timescales [Chontos et al., 2022]. Lomb–Scargle methods are useful for uneven sampling, but their statistical and window-function interpretation must be handled carefully [VanderPlas, 2018].

An échelle diagram folds the spectrum modulo a trial $\Delta\nu$. When the trial separation is correct, modes of the same degree align into nearly vertical ridges. The diagram compresses an asymptotic relation into a visually recognizable pattern and is routinely used to validate $\Delta\nu$, identify degrees, and inspect ridge curvature. However, ordinary plotting treats the horizontal coordinate as if it were an interval, even though it is intrinsically periodic. This proposal instead models the échelle diagram on a cylinder and studies the persistence of its ridges.

2.3. Persistent homology and stable topological summaries

Persistent homology studies a nested family of spaces or complexes

$$K_{a_1} \subseteq K_{a_2} \subseteq \cdots \subseteq K_{a_m}, \tag{2.2}$$

and records the parameter values at which homology classes appear and disappear [Edelsbrunner et al., 2002, Zomorodian and Carlsson, 2005]. A persistence diagram is a multiset of birth–death pairs (b_i, d_i). Long intervals are often interpreted as robust multiscale features, although physical relevance must be established by the application rather than inferred from lifetime alone.

For sublevel-set persistence of tame functions, the stability theorem states

$$d_{\mathrm{B}}(\mathrm{Dgm}_k(f), \mathrm{Dgm}_k(g)) \leq \|f - g\|_\infty , \tag{2.3}$$

under standard hypotheses [Cohen-Steiner et al., 2007]. Analogous interleaving and metric stability results exist for point-cloud filtrations. Persistence landscapes and images map diagrams into vector spaces suitable for statistics and machine learning [Bubenik, 2015, Adams et al., 2017]. Zigzag persistence permits sequences with both forward and backward maps and is relevant when structures are followed across windows or radial orders [Carlsson and de Silva, 2010].

2.4. Topology of time series and periodic signals

Takens-type delay embeddings reconstruct the geometry of an underlying dynamical system from scalar observations [Takens, 1981]. Sliding-window persistent homology makes this construction computational and relates prominent one-dimensional persistence to periodic content. For appropriate windows and embedding dimensions, periodic functions generate loop-like point clouds, while multiple Fourier components can produce higher-dimensional geometry [Perea and Harer, 2015]. These results motivate a mode-coherence analysis of stellar light curves.

The analogy must be used cautiously. A solar-like oscillation spectrum is not a small deterministic quasiperiodic system: modes are stochastically re-excited, damped, overlapping, and observed through granulation and an instrumental window. The relevant research question is therefore whether persistence can be calibrated against mode lifetime and coherence in a stochastic setting, not whether a visually appealing loop can be produced.

2.5. TDA in astronomy and adjacent work

Persistent homology has been used in cosmology to describe multiscale structure and, in some settings, to add constraining power beyond peak counts or lower-order summaries [Heydenreich et al., 2021]. A recent TDA application to Kepler light curves uses delay embeddings and topological features for exoplanet-transit classification [Chung, 2025]. These studies demonstrate computational feasibility but do not establish an asteroseismic mapping between persistence and solar-like mode parameters.

The present proposal is distinguished by four constraints:

1. every primary topological statistic is tied to a specific seismic observable or reliability question;
2. the natural periodic geometry of the échelle coordinate is retained;
3. simulations and published frequencies provide ground truth;
4. conditional-information and surrogate tests determine whether topology adds information beyond the PSD.

2.6. Research gap

The literature establishes the physical meaning of oscillation spectra, the statistical utility of persistent homology, and the feasibility of TDA in astronomy. It does not yet provide:

- a standard persistent-homology formulation of the échelle diagram on $S^1 \times I$;
- a validated topological estimator of $\Delta\nu$ or $\nu_{\max}$ for solar-like oscillators;
- an explicit link between angular ridge geometry and small frequency-separation ratios within a persistent framework;
- a calibration between delay-embedding persistence and stochastic mode lifetime or linewidth;
- a propagation of TDA stability bounds into uncertainty on stellar mass, radius, density, or gravity;
- a population-level comparison of classical and topological asteroseismic inference.

Research gap in one sentence

There is currently no physics-validated framework that treats the oscillation spectrum and échelle pattern of a solar-like star as multiscale topological objects and tests whether their persistent structure improves or certifies conventional asteroseismic inference.

3. Research questions, hypotheses, objectives, and scope

3.1. Primary research question

RQ0. Can persistent topology provide stable, physically interpretable, and statistically complementary observables of solar-like stellar oscillations, particularly under limited duration, irregular sampling, or reduced signal-to-noise?

3.2. Specific research questions

RQ1. Which data representation – light curve, delay embedding, PSD, mode-frequency point cloud, or échelle cylinder – produces the most reproducible and physically interpretable persistence?

RQ2. Can upper-star persistence of a background-normalized PSD recover $\nu_{\max}$ or a robust hierarchy of oscillation peaks?

RQ3. Can persistent ridge connectivity on the échelle cylinder estimate $\Delta\nu$ and reject common aliases such as $\Delta\nu/2$ or $2\Delta\nu$?

RQ4. Can angular ridge separation recover $\delta\nu_{02}$, r_{02}, r_{01}, and r_{10} with calibrated uncertainty?

RQ5. Does delay-embedding persistence correlate with mode amplitude, linewidth, lifetime, or quality factor after conditioning on the PSD?

RQ6. How do photometric perturbations, background uncertainty, gaps, cadence, and window functions alter persistence diagrams and derived stellar parameters?

RQ7. Do classical-plus-topological models improve held-out parameter recovery, posterior calibration, or automated quality control relative to strong classical baselines?

RQ8. Which results transfer from Kepler to TESS, and which are limited by duration, resolution, or contamination?

RQ9. Can analytical formulas or upper/lower bounds be obtained for idealized sinusoidal, multi-mode, and noisy mode-comb models?

3.3. Hypotheses and falsification criteria

ID	Hypothesis	Falsification / failure evidence
H1	PSD persistence supplies a stable estimator of the location and strength of the oscillation envelope.	Bias or variance consistently exceeds pySYD/Lightkurve without compensating gains in robustness or quality assessment.

ID	Hypothesis	Falsification / failure evidence
H2	Ridge persistence on the échelle cylinder is maximized near the physically correct $\Delta\nu$.	The score is dominated by smoothing, aliases, arbitrary graph parameters, or radial-order range across simulations and LEGACY stars.
H3	Persistent ridge geometry permits stable estimation of at least one small-separation ratio.	Angular ridge recovery fails to match peak-bagged ratios within calibrated uncertainty on adequate-S/N stars.
H4	Delay persistence contains information about coherence or damping.	Phase-randomized surrogates preserving the PSD reproduce the observed persistence distribution.
H5	Hybrid models improve inference or reliability.	Nested cross-validation shows no practically meaningful improvement after classical features are included.
H6	Stability theory yields useful astrophysical uncertainty bounds.	Bounds are formally true but too loose to inform any realistic observing or preprocessing decision.

3.4. General objective

To develop, mathematically characterize, and empirically validate a topological framework for extracting and certifying asteroseismic information from solar-like stellar light curves and oscillation spectra.

3.5. Specific objectives

O1. Reproduce conventional global and detailed seismic benchmarks for KIC 10963065.

O2. Construct and compare five physically motivated topological representations.

O3. Define topological estimators of $\nu_{\max}$ and $\Delta\nu$ with bootstrap or posterior uncertainty.

O4. Develop the cylindrical topological échelle diagram and ridge-persistence score.

O5. Test geometric/topological estimators of small-separation ratios.

O6. Derive exact relations or bounds for idealized signals and perturbations.

O7. Establish simulation-based operating regions in duration, S/N, cadence, and duty cycle.

O8. Validate on the 66-star Kepler LEGACY sample and matched control systems.

O9. Evaluate cross-mission generalization using TESS observations and Kepler-to-TESS degradation.

O10. Release reproducible code, data manifests, configurations, tests, and reference outputs.

3.6. Scope and exclusions

The core population comprises main-sequence and early subgiant solar-like oscillators for which p-mode ridges are measurable. Red giants may be included as a later stress test because mixed modes introduce qualitatively different ridge topology. The project does not initially aim to replace full Bayesian peak bagging, perform a uniform reanalysis of all Kepler stars, or infer stellar evolution parameters without conventional models. Topological deep learning

and multiparameter persistence are reserved for extension work unless simpler interpretable methods reach a clear performance ceiling.

3.7. Project architecture

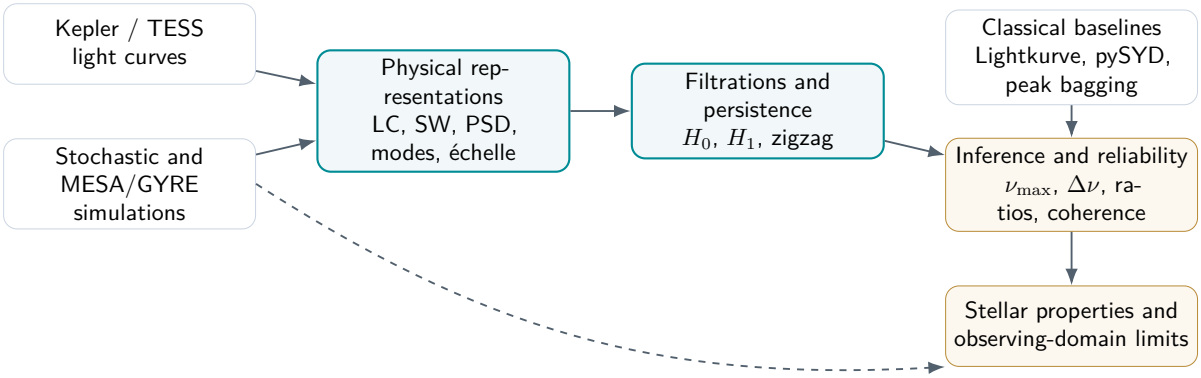


Figure 3.1.: Research architecture. Simulations and observations feed the same representation layer; topological quantities are compared with strong classical baselines before any physical claim is made.

4. Mathematical and physical framework

4.1. Solar-like oscillations as a physical inference problem

Solar-like oscillations are global acoustic modes stochastically excited and damped by near-surface convection. Their information is therefore distributed across at least three linked objects: a noisy and gapped flux time series, a power-density spectrum, and a structured family of eigenfrequencies. The proposed topology is not intended to bypass this physics. It is designed to encode structure at each of these levels and to determine which structures are stable enough to be interpreted.

4.1.1. Linear adiabatic oscillations

Let an equilibrium stellar model have density $\rho_0(r)$, pressure $p_0(r)$, gravitational potential $\Phi_0(r)$, and displacement field $\boldsymbol{\xi}(\boldsymbol{r}, t)$. Linearization of mass conservation, momentum conservation, Poisson's equation, and the adiabatic relation gives

$$\rho' + \nabla\cdot(\rho_0\boldsymbol{\xi}) = 0, \tag{4.1}$$

$$-\omega^2\rho_0\boldsymbol{\xi} = -\nabla p' - \rho'\nabla\Phi_0 - \rho_0\nabla\Phi', \tag{4.2}$$

$$\nabla^2\Phi' = 4\pi G\rho', \tag{4.3}$$

$$\frac{p'}{p_0} + \boldsymbol{\xi}\cdot\nabla\ln p_0 = \Gamma_1\left(\frac{\rho'}{\rho_0} + \boldsymbol{\xi}\cdot\nabla\ln\rho_0\right). \tag{4.4}$$

For a non-rotating spherical star, separation in spherical harmonics yields modes indexed by radial order n, angular degree ℓ, and azimuthal order m,

$$\boldsymbol{\xi}_{n\ell m}(\boldsymbol{r}, t) = [\xi_r(r)Y_\ell^m\hat{\boldsymbol{r}} + \xi_h(r)\nabla_h Y_\ell^m]\, e^{-i\omega_{n\ell m}t}. \tag{4.5}$$

Disk integration strongly favors low degrees, normally $\ell = 0, 1, 2, 3$. Rotation, structural asphericity, activity, and line-of-sight inclination perturb the ideal degeneracy in m. These effects will be retained in simulations when testing ridge geometry but will first be suppressed in proofs and unit tests.

4.1.2. Asymptotic p-mode pattern

For high radial order and low degree, the leading asymptotic relation is [Tassoul, 1980, Aerts et al., 2010]

$$\nu_{n\ell} \simeq \Delta\nu\left(n + \frac{\ell}{2} + \epsilon\right) - \ell(\ell+1)D_0 + \delta_{n\ell}^{\mathrm{curv}} + \delta_{n\ell}^{\mathrm{glitch}}, \tag{4.6}$$

where

$$\Delta\nu \simeq \left(2\int_0^R \frac{\mathrm{d}r}{c_s(r)}\right)^{-1} \propto \sqrt{\frac{M}{R^3}} \tag{4.7}$$

is the inverse acoustic diameter. The small separations

$$\delta\nu_{02}(n) = \nu_{n,0} - \nu_{n-1,2}, \tag{4.8}$$

$$\delta\nu_{01}(n) = \nu_{n,0} - \frac{\nu_{n-1,1} + \nu_{n,1}}{2} \tag{4.9}$$

probe departures from uniform acoustic spacing and are sensitive to core structure. Dimensionless ratios such as $r_{02} = \delta\nu_{02}/\Delta\nu$ reduce sensitivity to poorly modeled near-surface layers. In an ordinary échelle diagram, $x = \nu \bmod \Delta\nu$ and $y = \nu$ (or radial order); the asymptotic comb becomes a small set of approximately vertical ridges.

4.1.3. Stochastic excitation, damping, and background

A mode coordinate can be modeled locally as a stochastically forced damped oscillator,

$$\ddot{q} + 2\eta\dot{q} + \omega_0^2 q = f(t), \tag{4.10}$$

whose expected power spectral density (PSD) near resonance is approximately Lorentzian,

$$P_{n\ell m}(\nu) = \frac{H_{n\ell m}}{1 + 4(\nu - \nu_{n\ell m})^2/\Gamma_{n\ell m}^2}, \qquad \tau_{n\ell m} \simeq \frac{1}{\pi\Gamma_{n\ell m}}. \tag{4.11}$$

The oscillation envelope is represented at first order by a Gaussian centered on $\nu_{\max}$, while the convective background is modeled as a sum of generalized Harvey components [Harvey, 1985, Kallinger et al., 2014],

$$B(\nu) = W + \sum_{j=1}^{J} \frac{4\sigma_j^2\tau_j}{1 + (2\pi\nu\tau_j)^{\alpha_j}}. \tag{4.12}$$

The observation model multiplies the underlying signal by a sampling window and adds measurement noise. This point is essential: topology computed from an observed periodogram is topology of a convolved, finite-duration estimator, not directly topology of the stellar PSD.

4.1.4. Scaling relations and propagated uncertainty

Using solar reference values consistently calibrated to the selected pipeline, the standard relations are [Kjeldsen and Bedding, 1995, Chaplin and Miglio, 2013]

$$\frac{R}{R_\odot} \simeq \left(\frac{\nu_{\max}}{\nu_{\max,\odot}}\right)\left(\frac{\Delta\nu}{\Delta\nu_\odot}\right)^{-2}\left(\frac{T_{\rm eff}}{T_{\rm eff,\odot}}\right)^{1/2}, \tag{4.13}$$

$$\frac{M}{M_\odot} \simeq \left(\frac{\nu_{\max}}{\nu_{\max,\odot}}\right)^{3}\left(\frac{\Delta\nu}{\Delta\nu_\odot}\right)^{-4}\left(\frac{T_{\rm eff}}{T_{\rm eff,\odot}}\right)^{3/2}, \tag{4.14}$$

$$\frac{g}{g_\odot} \simeq \left(\frac{\nu_{\max}}{\nu_{\max,\odot}}\right)\left(\frac{T_{\rm eff}}{T_{\rm eff,\odot}}\right)^{1/2}. \tag{4.15}$$

For small relative errors and covariance matrix Σ in $\boldsymbol{z} = (\ln\nu_{\max}, \ln\Delta\nu, \ln T_{\rm eff})$, the covariance of any log-derived quantity is $J\Sigma J^{\mathsf{T}}$. For example,

$$\mathrm{Var}(\ln M) = \boldsymbol{a}_M^{\mathsf{T}}\Sigma\boldsymbol{a}_M, \qquad \boldsymbol{a}_M = (3, -4, 3/2)^{\mathsf{T}}. \tag{4.16}$$

This makes the physical cost of topological estimator errors explicit and defines practical accuracy targets.

4.2. Persistent homology and stable summaries

4.2.1. Filtered spaces and persistence diagrams

Given a filtration $X_a \subseteq X_b$ for $a \le b$, homology tracks connected components, loops, and higher-dimensional voids through the maps $H_k(X_a) \to H_k(X_b)$. A feature born at b_i and dying at d_i is recorded as a point (b_i, d_i) in the persistence diagram

$$\mathrm{Dgm}_k(X) = \{(b_i, d_i) : b_i < d_i\}, \qquad \pi_i = d_i - b_i. \tag{4.17}$$

The project will principally use H_0 for peak or basin organization and H_1 for recurrent phase geometry. Vietoris–Rips filtrations will be used for point clouds; sublevel/superlevel filtrations for scalar functions; cubical filtrations for images; and zigzag persistence where frequency-order slices enter and leave a sequence.

For two diagrams D and E, the bottleneck distance is

$$d_{\mathrm{B}}(D,E) = \inf_{\gamma} \sup_{x \in D} \|x - \gamma(x)\|_{\infty}\,, \tag{4.18}$$

where matching to the diagonal is permitted. Wasserstein distances, persistence landscapes [Bubenik, 2015], images [Adams et al., 2017], entropy, and Betti curves will be secondary summaries. Diagram-level inference will be preferred whenever feasible so that vectorization does not conceal which features drive a result.

4.2.2. Stability as an astrophysical quality guarantee

For tame functions $f, g : X \to \mathbb{R}$, sublevel-set persistence obeys [Cohen-Steiner et al., 2007]

$$d_{\mathrm{B}}(\mathrm{Dgm}_k(f), \mathrm{Dgm}_k(g)) \le \|f - g\|_{\infty}\,. \tag{4.19}$$

Thus a flux or power perturbation bounded by ε cannot move any persistence point by more than ε under the appropriate filtration. This is a rigorous robustness statement, but it becomes scientifically useful only after (i) the normalization has physical units, (ii) detrending error is included, and (iii) the bound is compared with the persistence margin separating a claimed stellar feature from the diagonal.

Physical interpretation rule

No persistence class will be labeled "stellar" solely because it is long-lived. A class must also be localized in a physically relevant frequency or time domain, survive the observation model, exceed calibrated surrogate/null distributions, and behave consistently under controlled changes of preprocessing.

4.3. Five complementary topological constructions

4.3.1. A. Sublevel and superlevel topology of the light curve

Let $x(t_i)$ be normalized relative flux. The lower-star filtration $X_a = \{t : x(t) \le a\}$ records the merging of flux minima; applying it to $-x$ records maxima. In one dimension, H_0 persistence pairs local extrema and is closely related to prominence. This representation is inexpensive and useful for activity, flares, instrumental discontinuities, and preprocessing diagnostics. It is not expected to resolve individual p modes directly because many modes superpose in flux.

4.3.2. B. Sliding-window or delay-coordinate geometry

For delay τ and embedding dimension m, define

$$\mathrm{SW}_{m,\tau}\, x(t) = (x(t), x(t+\tau), \ldots, x(t+(m-1)\tau)) \in \mathbb{R}^m. \tag{4.20}$$

After centering and, where justified, sphere normalization, a coherent oscillation traces a loop; amplitude modulation thickens or radially deforms it; multiple incommensurate components suggest a toroidal geometry. Persistent H_1 quantifies the loop without requiring a specific phase origin [Takens, 1981, Perea and Harer, 2015]. The delay will be selected using the dominant period and mutual-information/autocorrelation diagnostics, while m will be checked by persistence convergence rather than a single embedding heuristic.

For an ideal sinusoid $x(t) = A\cos(\omega t + \phi)$, every delay vector lies in the two-dimensional span of

$$\boldsymbol{u} = (1, \cos\omega\tau, \ldots, \cos((m-1)\omega\tau)), \tag{4.21}$$
$$\boldsymbol{v} = (0, \sin\omega\tau, \ldots, \sin((m-1)\omega\tau)), \tag{4.22}$$

and therefore traces an ellipse $A(\cos\theta\,\boldsymbol{u} - \sin\theta\,\boldsymbol{v})$. If $\boldsymbol{u}$ and $\boldsymbol{v}$ are orthogonal with equal norms, the ellipse is a circle. This gives an exact bridge from frequency and amplitude to embedded geometry; the research question is how damping, stochastic re-excitation, gaps, and multiple modes deform its persistence.

4.3.3. C. Superlevel topology of the background-normalized PSD

Let $S(\nu) = P(\nu)/B(\nu)$ or $\log P(\nu) - \log B(\nu)$. For the upper-star filtration $U_a = \{\nu : S(\nu) \geq a\}$, a peak is born at its local maximum and dies when it merges into an older component at the relevant saddle/minimum. In one dimension, its lifetime equals its topographic prominence under the elder rule. Consequently, PSD H_0 is not merely a generic feature extractor: it is an exact multiscale representation of peak prominence.

Define persistence-weighted peak locations ν_i and lifetimes π_i after excluding the low-frequency activity domain. Candidate global observables are

$$\widehat{\nu}^{\mathrm{top}}_{\mathrm{max}} = \frac{\sum_i w_i \nu_i}{\sum_i w_i}, \qquad w_i = \pi_i^q K_h(\nu_i - \widetilde{\nu}), \tag{4.23}$$

$$C_{\mathrm{comb}}(\Delta) = \sum_{i<j} w_i w_j \sum_{k=1}^{K} \exp\left[-\frac{(|\nu_i - \nu_j| - k\Delta/2)^2}{2s^2}\right], \tag{4.24}$$

where K_h is a robust envelope kernel and q, h, s, K are fixed using simulations, not tuned per star. The maximizer over a physically constrained interval provides one candidate $\widehat{\Delta\nu}^{\mathrm{top}}$.

4.3.4. D. Topology and geometry of extracted mode frequencies

When a peak list or posterior sample of mode frequencies is available, each frequency is mapped to

$$\boldsymbol{z}_i(\Delta) = \left(\cos\frac{2\pi\nu_i}{\Delta}, \sin\frac{2\pi\nu_i}{\Delta}, \lambda\frac{\nu_i - \nu_c}{\Delta}\right). \tag{4.25}$$

For the correct trial Δ, modes of a given degree occupy nearly longitudinal strands on a cylinder. Persistence quantifies whether a coherent circular phase organization exists; graph geometry supplies ridge separation and curvature. Posterior samples propagate frequency uncertainty into distributions over diagrams rather than a single brittle point cloud.

4.3.5. E. The topological échelle diagram

For a trial spacing Δ, define the cylindrical map

$$E_\Delta : \nu \mapsto (q, \theta) = \left(\left\lfloor\frac{\nu - \nu_0}{\Delta}\right\rfloor, 2\pi\,\mathrm{frac}\,\frac{\nu - \nu_0}{\Delta}\right), \qquad \theta \in S^1. \tag{4.26}$$

The spectrum is accumulated into a periodic image $I_\Delta(q, \theta)$, background-normalized, and filtered cubically. A ridge graph is also constructed from high-persistence local maxima, with periodic angular distance

$$d_{S^1}(\theta, \theta') = \min_{k\in\mathbb{Z}} |\theta - \theta' + 2\pi k|. \tag{4.27}$$

The proposed score combines ridge survival, angular concentration, continuity across orders, and a complexity penalty:

$$\mathcal{T}(\Delta) = \sum_{r \in \mathcal{R}_\Delta} L_r \, \bar{\pi}_r \, \exp(-\sigma_{\theta,r}^2/2) - \lambda_1 N_{\text{break}} - \lambda_2 N_{\text{spurious}}. \tag{4.28}$$

Here L_r is ridge length in radial orders and $\bar{\pi}_r$ its mean prominence/persistence. The estimator is $\widehat{\Delta\nu}^{\text{top}} = \arg\max_{\Delta \in \mathcal{I}} \mathcal{T}(\Delta)$, with uncertainty obtained by block bootstrap, posterior predictive simulation, or both. This is a genuinely topological échelle analysis only if the score adds stability or information beyond image contrast and conventional ridge correlation; that distinction will be tested explicitly.

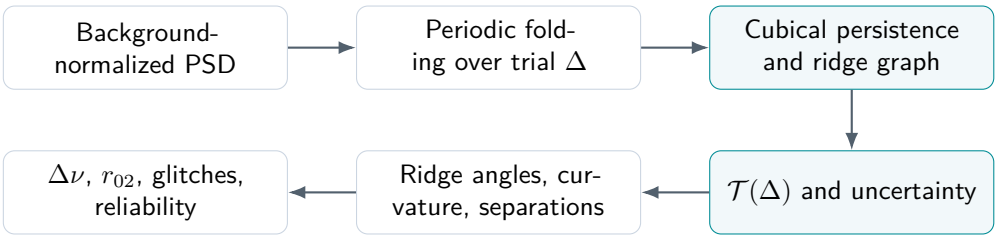


Figure 4.1.: Computational definition of the topological échelle diagram. Periodicity is respected by the cylindrical metric; persistent structures are converted to physical quantities only after null calibration.

4.4. Analytical relationships and theorem targets

The project distinguishes established results, straightforward propositions to be proved in the thesis, and conjectures requiring empirical validation.

Table 4.1.: Planned mathematical results and their astrophysical use.

Result	**Mathematical statement**	**Astrophysical interpretation**
Peak–persistence equivalence	For a generic one-dimensional scalar function, upper-star H_0 lifetime is the peak prominence defined by the elder-rule merge tree.	Converts PSD peak morphology into an exact topological quantity; no machine learning is required.
Sinusoidal loop proposition	The sliding-window image of a sinusoid lies on an ellipse in a two-dimensional subspace; its axes follow from the Gram matrix of $\boldsymbol{u}, \boldsymbol{v}$.	Links amplitude, frequency, delay choice, and H_1 detectability.
Perturbation bound	If normalized spectra obey $\left\|S - \widetilde{S}\right\|_\infty \leq \varepsilon$, then their diagrams differ by at most ε in bottleneck distance.	A persistence margin $> 2\varepsilon$ certifies that a feature cannot be erased by that perturbation alone.
Fold-phase bound	$d_{S^1}(2\pi\nu/\Delta, 2\pi\widetilde{\nu}/\Delta) \leq 2\pi\lvert\nu - \widetilde{\nu}\rvert/\Delta$.	Propagates frequency posterior error into ridge-angle uncertainty.

Result	Mathematical statement	Astrophysical interpretation
Small-separation identity	If degree labels are correct, $r_{02}(n) = d_{S^1}(\theta_{n,0}, \theta_{n-1,2})/(2\pi)$ up to orientation and order convention.	Makes a standard seismic ratio a normalized angular separation on the échelle cylinder.
Spacing-mismatch drift	For N consecutive orders and trial $\Delta = \Delta\nu + \delta$, ideal ridge drift is $O(N\lvert\delta\rvert/\Delta\nu)$ modulo curvature.	Predicts the width of the maximum of $\mathcal{T}(\Delta)$ and its dependence on observing bandwidth.
Scaling bound	Relative bounds on $(\hat{\nu}^{\mathrm{top}}_{\max}, \widehat{\Delta\nu}^{\mathrm{top}}, T_{\mathrm{eff}})$ induce first-order bounds on M, R, g through equation (4.16).	Converts topology-level stability into a physical error budget.

4.4.1. A concrete lower bound on ridge persistence

Suppose an ideal ridge has intensity at least a over L adjacent orders, while every path connecting it to a competing ridge crosses intensity no greater than $b < a$. In the superlevel merge tree, the corresponding component has persistence at least $a - b$. If an observed image differs uniformly by at most ε, stability guarantees a matched class with persistence at least

$$\pi_{\mathrm{obs}} \geq (a - b) - 2\varepsilon. \tag{4.29}$$

This simple inequality is a prototype of the desired physical bounds: the contrast margin, background uncertainty, and ridge survival appear in a common expression. Tighter bounds incorporating correlated spectral noise and periodic cubical complexes constitute a thesis-level mathematical contribution.

4.5. Interpretive limits

Homology discards metric and label information unless these are reintroduced carefully. Two spectra can have similar diagrams but very different frequency locations; conversely, small spectral perturbations may substantially change an estimator that includes discontinuous peak selection. Therefore, the project will pair topology with localization, circular geometry, and probabilistic labeling. The word “equivalence” will be reserved for proven identities such as peak prominence or idealized phase separation; most links between persistence and stellar parameters will be calibrated relations with stated domains of validity.

5. Data, experiments, and computational methodology

5.1. Study design

The project uses a ladder of increasing realism. Mathematical unit cases establish what each construction measures; controlled simulations establish operating limits; Rudy provides a transparent end-to-end benchmark; the Kepler LEGACY stars test population validity; and TESS tests transfer under a different cadence, baseline, aperture, and noise regime. At every rung, a classical method and a topology-free ablation are retained.

Table 5.1.: Data tiers and inferential roles. Catalog holdings will be frozen in a versioned manifest at project start.

Tier	Material	Purpose	Primary output
T0: analytic	Sinusoids, combs, Lorentzians, ideal ridges	Proofs, invariance tests, exact unit tests	Verified identities and implementation tests
T1: stochastic synthetic	Solar-like mode/background simulator; controlled windows, S/N, damping, cadence	Bias, variance, detection boundaries, null calibration	Operating-domain maps
T2: stellar models	MESA structures and GYRE frequencies across a modest evolutionary grid	Physical ridge curvature, glitches, mixed-mode onset	Model-to-topology response surfaces
T3: benchmark	KIC 10963065 short-cadence Kepler quarters 2, 5, 6, 7; TESS target data where usable	Full audit trail and visual demonstration	Reproducible Rudy case study
T4: validation	66-star Kepler LEGACY sample with published frequencies/properties	External labels, population generalization, calibration	Comparative catalog and performance paper
T5: stress tests	Matched solar analogs/subgiants; Kepler degraded to TESS-like windows; actual TESS sectors	Cross-mission robustness and failure analysis	Transfer limits and quality flags

5.2. Benchmark star: KIC 10963065 (Rudy)

Rudy is an advantageous case because it is a solar-like dwarf with public short-cadence Kepler data and an established tutorial-scale analysis. The Lightkurve reference workflow uses quarters 2, 5, 6, and 7 and reports approximately $\nu_{\max} = 2145\,\mu$Hz, $\Delta\nu = 103.11\,\mu$Hz, $M \simeq 1.08 M_\odot$, $R \simeq 1.23 R_\odot$, and $\log g \simeq 4.29$ [Hall and Barentsen, 2020]. These numbers are reproducibility

targets, not immutable truths; posterior results from independent background fits and pySYD will define the reference range.

The star is cross-identified as TIC 48450369 in the MAST holdings inspected for this proposal. A proposal-stage MAST query in July 2026 returned SPOC target light-curve holdings in TESS sectors 14, 15, 40, 41, 53, 55, 74, 75, 80, 81, and 82 [Mikulski Archive for Space Telescopes, 2026]. This list will be re-queried and frozen in the project manifest because archive holdings and product versions can change. The products will be analyzed by sector and jointly only after confirming cadence, aperture contamination, duty cycle, and whether the oscillation envelope is detectable. The expected limitation is not the Nyquist frequency for short-cadence products but the much shorter effective baseline and higher noise. TESS is therefore framed as a transfer experiment, not promised as a second precision benchmark.

5.3. Acquisition, provenance, and data quality

All target product identifiers, mission, sector/quarter, cadence, author, pipeline version, download timestamp, checksum, and quality-mask policy will be stored in a machine-readable manifest. Raw or minimally processed SAP flux and pipeline-corrected PDCSAP flux will be retained. No data file will be silently overwritten; each derived series will point to its parent and configuration hash.

Data quality will be reported before astrophysical interpretation:

1. time ordering, duplicate timestamps, cadence distribution, missing fraction, and contiguous-segment lengths;
2. finite-value and quality-flag counts before and after masking;
3. median flux, robust scatter, point-to-point scatter, and time-dependent variance;
4. spectral window, effective duty cycle, and principal alias peaks;
5. aperture/crowding metrics and evidence of quarter/sector discontinuities;
6. comparison of SAP and PDCSAP behavior in the oscillation and low-frequency bands.

Failures remain in an exclusion ledger with reason codes. This prevents a clean final sample from concealing selection effects.

5.4. Preprocessing and sensitivity branches

The primary branch will use mission quality flags, robust outlier filtering, quarter-wise normalization, and a high-pass correction with a cutoff far below the p-mode band. Because preprocessing can manufacture or erase topology, the following branches are mandatory:

- SAP versus PDCSAP and, where available, community-corrected products;
- conservative versus standard quality masks;
- spline, Savitzky–Golay, and Gaussian-process long-trend removal;
- no gap filling for Lomb–Scargle estimation versus explicitly labeled inpainting for FFT-based checks;
- transit/flare masks versus unmasked controls;
- concatenated versus segment-wise topology and spectra.

Filter timescales will be fixed from the physical band of interest and verified using transfer functions. Persistence diagrams will be compared across branches using bottleneck/Wasserstein

distances. A preprocessing decision is accepted only if recovered injected signals are unbiased and conclusions do not depend on one undocumented setting.

5.5. Classical asteroseismic baseline

The baseline is a scientific control, not a cosmetic comparison. It will include:

1. Lomb–Scargle or FFT power-density spectrum with correct normalization and window characterization [VanderPlas, 2018];
2. multi-component background fit using equation (4.12), with and without an oscillation-envelope component;
3. $\nu_{\max}$ from the smoothed background-divided excess;
4. $\Delta\nu$ from autocorrelation, power spectrum of the power spectrum, and pySYD where applicable [Chontos et al., 2022];
5. conventional échelle diagrams over a blind grid in Δ followed by a locked diagnostic plot;
6. peak bagging or published posterior mode frequencies for validation stars;
7. scaling-relation M, R, g with covariance-aware error propagation.

For Rudy, the Lightkurve tutorial is first reproduced line by line, then replaced with a version-pinned pipeline whose intermediate arrays are stored. Agreement will be evaluated against reported values and independent published/automated estimates, including the recent WhoS-GlAd analysis of Rudy and related bright solar-like stars [Özel et al., 2026].

5.6. Simulation program

5.6.1. Fast stochastic simulator

The basic simulator draws each mode from a celerite-style stochastic harmonic oscillator or an equivalent Fourier-domain Gaussian process, applies a Gaussian-like mode-height envelope, includes degree-dependent visibility and rotational multiplets, adds Harvey backgrounds and photon noise, and samples through an actual or synthetic window. A simulated observation is

$$y_i = w_i \left[\sum_{n\ell m} q_{n\ell m}(t_i) + g(t_i) + a(t_i) \right] + \epsilon_i, \tag{5.1}$$

where g is granulation, a activity/systematics, w_i the sampling mask, and ϵ_i heteroscedastic measurement noise. Inputs and random seeds will be recorded.

5.6.2. Experimental factors

The core factorial design spans:

- $\Delta\nu$ and $\nu_{\max}$ representative of dwarfs and early subgiants;
- observing duration from one TESS sector to multi-quarter and four-year Kepler baselines;
- cadence and finite integration attenuation;
- oscillation-to-background ratio, photon noise, and crowding dilution;
- mode linewidth/lifetime, envelope width, inclination, rotational splitting, and activity shifts;
- duty cycle, periodic gaps, random gaps, quarter boundaries, and momentum-dump-like interruptions;

- ridge curvature, acoustic glitches, surface terms, and onset of mixed modes.

A space-filling Latin hypercube will cover continuous factors; targeted grids will resolve expected detection boundaries. At least 200 replicates per boundary cell will be generated for reliable detection and coverage estimates, while broad model-comparison experiments will use sequential Monte Carlo error monitoring to stop once uncertainty in the performance contrast is below a prespecified tolerance.

5.6.3. Physics-informed stellar models

A compact MESA grid [Paxton et al., 2011] will span mass, metallicity, helium, age/evolutionary state, and mixing assumptions around the benchmark population. GYRE [Townsend and Teitler, 2013] supplies adiabatic low-degree frequencies. This tier isolates structural changes in ridge topology without realization noise. Surface corrections and stochastic observation models are added in separate layers so their effects can be identified rather than conflated.

5.7. Topology pipeline

5.7.1. Representation-specific computation

The implementation will use Ripser/GUDHI/giotto-tda as independently testable backends where their capabilities overlap [Bauer, 2021, Tauzin et al., 2021]. Each representation has a declared metric, filtration, homology degree, coefficient field, truncation threshold, and normalization. Default coefficient field is $\mathbb{F}_2$; any alternative must be motivated.

Table 5.2.: Primary topological representations and controls.

Object	**Filtration**	**Feature**	**Physical target**	**Critical control**
Flux series	lower/upper-star	H_0 extrema	activity/systematics; preprocessing QC	shuffled and block-surrogate flux
Delay cloud	Vietoris–Rips	dominant H_1, landscape	phase coherence; damping proxy	IAAFT and phase-randomized surrogates
Normalized PSD	upper-star/merge tree	H_0 lifetime and locations	peak prominence, envelope, comb	background-only and injected spectra
Mode cloud	Rips on cylinder	strands/loops plus graph geometry	spacing and mode phase	randomized frequencies with same density
Échelle image	periodic cubical	components, loops, zigzag lifetime	ridge survival, $\Delta\nu$, ratios	topology-free contrast/ridge score

5.7.2. Algorithm for the topological échelle estimator

Algorithm 1: Cylindrical topological échelle analysis

Input: frequencies or normalized spectrum $S(\nu)$; trial interval $\mathcal{I}$; bootstrap count B.

1. For every $\Delta \in \mathcal{I}$, map ν to (q, θ) using equation (4.26) and enforce periodic angular adjacency.
2. Construct $I_\Delta(q, \theta)$ and its superlevel cubical filtration.
3. Compute persistence diagrams and localize high-persistence components.
4. Link maxima between adjacent orders using circular distance and uncertainty.
5. Compute $\mathcal{T}(\Delta)$ from ridge lifetime, continuity, concentration, and penalties.
6. Set $\widehat{\Delta\nu}^{\mathrm{top}} = \arg\max_{\Delta \in \mathcal{I}} \mathcal{T}(\Delta)$.
7. For $b = 1, \ldots, B$, resample realization-noise blocks or draw a posterior predictive spectrum and repeat the complete selection.
8. Report the estimator, interval, competing maxima, ridge graph, diagram, and reliability diagnostics.

5.7.3. Parameter selection without leakage

Filtration thresholds are intrinsic to persistence, but preprocessing bandwidths, embedding parameters, diagram kernels, and ridge-linking penalties remain hyperparameters. They will be selected on simulations and training folds only. For population studies, all measurements from the same star remain in one fold. TESS sectors for a Kepler star never enter a validation fold if its Kepler data were used for tuning. Hyperparameters and decision thresholds will be frozen before opening the final LEGACY test partition.

5.8. Surrogate tests and null hypotheses

False topological structure can arise from colored noise, the spectral window, smoothing, or deterministic detrending. Four nested nulls will be used:

1. parametric Harvey-background plus photon-noise simulations with the real sampling window;
2. Fourier phase-randomized surrogates preserving the periodogram while destroying phase relations [Theiler et al., 1992];
3. iterative amplitude-adjusted Fourier transform (IAAFT) surrogates preserving the marginal distribution and approximate PSD [Schreiber and Schmitz, 1996];
4. injected-signal controls in which known p-mode patterns are added to observed quiet/background segments.

For a topological statistic T, an empirical p-value is

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{I}(T_b \geq T_{\mathrm{obs}})}{B + 1}. \tag{5.2}$$

Multiple testing across trial spacings or frequency bands is controlled by using the maximum statistic in every surrogate, or by false-discovery-rate control for explicitly separate hypotheses. Effect sizes and detection probabilities, not only p-values, will be reported.

5.9. Inference, uncertainty, and comparison metrics

5.9.1. Uncertainty propagation

Three levels are distinguished:

- **realization uncertainty**: block bootstrap or posterior predictive draws consistent with stochastic modes and correlated background;
- **measurement/model uncertainty**: posterior draws of background, mode frequencies, and nuisance parameters;
- **pipeline uncertainty**: spread across justified preprocessing branches and hyperparameter sensitivity.

Confidence or credible intervals must rerun feature construction and selection. Holding a selected ridge fixed while resampling would underestimate uncertainty.

5.9.2. Performance metrics

For a quantity $\theta \in \{\nu_{\max}, \Delta\nu, r_{02}, M, R, g\}$, comparison uses relative bias, median absolute error, RMSE, normalized residuals, 68/95% interval coverage, interval width, and catastrophic-failure rate. Detection tasks use sensitivity, specificity, area under precision–recall curves, and false-alarm probability at fixed completeness. Reliability scores are evaluated with calibration curves and Brier/log scores.

The key complementary-value test is incremental predictive information. If X_c denotes classical features and X_t topological features, then a hybrid result is scientifically positive only when nested cross-validation shows a reproducible improvement in one or more locked quantities such as

$$\Delta R^2 = R^2(X_c, X_t) - R^2(X_c), \qquad \Delta \log p = \log p(y \mid X_c, X_t) - \log p(y \mid X_c), \tag{5.3}$$

with uncertainty excluding a practically negligible interval. Conditional permutation importance and conditional mutual information will check whether the gain is not a disguised restatement of $\nu_{\max}$ or S/N.

5.9.3. Hybrid models

The first hybrid model is deliberately interpretable: regularized linear/GAM regression or hierarchical Bayesian calibration using classical estimates, selected persistence summaries, quality indicators, and interactions known from the simulations. Gradient-boosted trees form a nonlinear benchmark. Topological neural networks are outside the core scope unless sample size and ablations justify them. Stellar covariates and evolutionary state enter hierarchical terms so performance is not averaged across incompatible regimes.

5.10. Validation protocol and power

The 66-star LEGACY sample [Lund et al., 2017, Silva Aguirre et al., 2017] permits precise within-star measurements but only moderate population-level model complexity. Accordingly:

1. analytic relations and simulations carry the main identification burden;
2. population models remain low-dimensional and regularized;
3. repeated nested five-fold cross-validation is grouped by star;
4. a final holdout of approximately 20% is opened once after model/threshold lock;
5. paired bootstrap confidence intervals compare methods on the same stars;
6. no claim rests on subgroup results with fewer than a prespecified minimum number of stars.

For paired standardized performance differences, $N = 66$ has approximately 80% power near a moderate effect ($d \approx 0.35$) under ideal Gaussian assumptions; actual uncertainty will be evaluated by simulation using the observed error distribution. The project is therefore powered to detect useful complementary gains, not tiny universal improvements.

5.11. Ablations, robustness, and negative controls

Every major claim includes a topology-free counterpart:

- peak prominence versus the equivalent H_0 implementation;
- conventional autocorrelation/comb response versus equation (4.24);
- image ridge contrast and Hough-like continuity versus cubical persistence;
- ordinary circular statistics versus Rips persistence on the mode cloud;
- PSD-only classifiers versus delay-topology features;
- correct spacing versus half/double-spacing aliases and randomized mode labels.

An ablation that matches the full method is an informative negative result: it identifies which apparent gains come from representation or geometry rather than algebraic topology.

5.12. Software architecture and reproducibility

The project will be released as a versioned Python package and workflow rather than a collection of notebooks. Notebooks will serve only as readable demonstrations.

Listing 5.1: Planned reproducible repository structure.

```
topological-asteroseismology/
  pyproject.toml              # locked dependencies and package
      metadata
  configs/                    # immutable YAML experiment
      configurations
  data/manifests/             # MAST identifiers, checksums, provenance
  src/topoastro/              # preprocessing, spectra, topology,
      inference
  tests/                      # analytic, regression, and property-
      based tests
  workflows/                  # download, simulate, fit, validate,
      publish
  notebooks/                  # Rudy and tutorial-facing demonstrations
  results/                    # machine-readable tables and figure
      recipes
  environment/                # conda/uv lock and container definition
```

Continuous integration will test exact sinusoidal/comb cases, invariance under time/phase translation where appropriate, persistence stability under bounded perturbations, and numerical agreement between independent backends. Random seeds, package versions, CPU/GPU information, and configuration hashes accompany each published result. MAST data are referenced by manifest rather than redistributed when licensing or volume makes redistribution unnecessary.

5.13. Computational resources

The baseline and one-dimensional persistence computations run on a workstation. Simulation ensembles, cubical persistence grids, MESA/GYRE models, and nested bootstrap/cross-validation

are parallel batch workloads. A planning allocation of 50,000–100,000 CPU-hours, 2–5 TB temporary storage, and under 200 GB archival derived products is conservative; pilot profiling will replace these figures in month 3. GPU resources are not required for the core thesis.

6. Work packages, deliverables, and decision gates

6.1. Work-package logic

The work packages (WPs) are ordered so that the doctorate remains publishable even if the most ambitious construction fails. WP1–WP4 yield a methods/benchmark paper; WP5 provides the population test; WP6 contains the deeper mathematical and échelle contribution; and WP7 establishes complementarity and mission transfer. Each package has an exit criterion that prevents indefinite method tuning.

6.1.1. WP0 – Governance, literature map, and reproducibility foundation

Aim. Convert the proposal into a preregistered, traceable research program.

Tasks. Maintain a living systematic literature map; freeze terminology and data dictionaries; create the repository, testing policy, and archival plan; register the hypotheses, primary metrics, and holdout procedure; complete MAST data manifests; document computational-resource access.

Deliverables. D0.1 literature matrix and gap statement; D0.2 reproducibility plan; D0.3 tested repository skeleton; D0.4 locked benchmark protocol.

Exit criterion. A new collaborator can reproduce acquisition of Rudy and one LEGACY star from only the manifest and documentation.

6.1.2. WP1 – Classical benchmark and observation model

Aim. Establish a trusted conventional reference before TDA is introduced.

Tasks. Reproduce the Lightkurve example; run pySYD; fit backgrounds; compute the spectral window; build ordinary échelle diagrams; validate normalization and uncertainties; catalog Kepler/TESS holdings; compare sector/quarter products.

Deliverables. D1.1 Rudy benchmark notebook and pipeline; D1.2 baseline table with $\nu_{\max}$, $\Delta\nu$, M, R, g; D1.3 data-quality and window report; D1.4 TESS feasibility note.

Exit criterion. Global parameters agree with defensible reference ranges within joint uncertainty, or discrepancies are traced to documented modeling choices.

6.1.3. WP2 – Analytic cases and stochastic simulation laboratory

Aim. Determine what the topological constructions measure under known truth.

Tasks. Prove the sinusoidal embedding and peak-prominence results; implement comb/Lorentzian/ridge tests; create stochastic solar-like simulations; inject mission windows; generate surrogate families; construct initial detection-boundary maps.

Deliverables. D2.1 analytic test suite; D2.2 validated simulator; D2.3 simulation design and pilot archive; D2.4 false-alarm calibration.

Exit criterion. Nominal interval coverage and type-I error are achieved in locked pilot regimes, and software passes cross-backend regression tests.

6.1.4. WP3 – Representation comparison and estimator development

Aim. Select the smallest set of topological representations that offers distinct information.

Tasks. Implement constructions A–E from section 4.3; compare diagram summaries; quantify redundancy; develop $\hat{\nu}_{\max}^{\mathrm{top}}$, comb, and coherence statistics; perform topology-free ablations; select hyperparameters only from simulation training data.

Deliverables. D3.1 representation benchmark; D3.2 estimator API; D3.3 ablation report; D3.4 locked primary statistic set.

Exit criterion. Every retained statistic has an interpretable physical target, a calibrated null, and either accuracy or reliability value beyond a simpler control.

6.1.5. WP4 – Rudy end-to-end topological case study

Aim. Produce an auditable worked example suitable for the first publication.

Tasks. Compute all representations on Rudy; compare quarters; propagate preprocessing and background uncertainty; construct conventional and topological échelle diagrams; run surrogate tests; perform Kepler-to-TESS degradation; analyze actual TESS products if the envelope is detectable.

Deliverables. D4.1 complete Rudy analysis; D4.2 interactive/archival diagnostic figures; D4.3 reproducibility capsule; D4.4 first manuscript.

Exit criterion. All headline plots and tables can be regenerated by one command from public identifiers, and conclusions survive registered sensitivity branches.

6.1.6. WP5 – Population validation on Kepler LEGACY stars

Aim. Test generalization and identify the valid stellar/observational domain.

Tasks. Process the 66 stars; compare with published mode frequencies and properties; run grouped nested validation; measure calibration and failure rates; stratify by evolutionary state, magnitude, S/N, linewidth, and duty cycle; open the frozen holdout once.

Deliverables. D5.1 topology-enhanced LEGACY catalog; D5.2 operating-domain atlas; D5.3 second manuscript; D5.4 machine-readable benchmark dataset.

Exit criterion. Performance estimates have paired uncertainty and no data leakage; failure domains are stated as explicitly as successful domains.

6.1.7. WP6 – Theory and topological échelle formalization

Aim. Deliver the distinctive mathematical contribution.

Tasks. Formalize the periodic cubical complex; derive spacing-mismatch and perturbation bounds; study ridge zigzag persistence; connect ridge angle to separation ratios; test curvature/glitch sensitivity on MESA/GYRE grids; refine theorem assumptions and counterexamples.

Deliverables. D6.1 theorem/proposition chapter with proofs; D6.2 open topological échelle implementation; D6.3 model-grid response paper or methods section; D6.4 catalog of counterexamples.

Exit criterion. At least one nontrivial relation is proven and one estimator is empirically calibrated; otherwise the output is reframed honestly as a stability/representation study.

6.1.8. WP7 – Hybrid inference and TESS transfer

Aim. Determine whether TDA is genuinely complementary and mission-portable.

Tasks. Fit interpretable hybrid models; evaluate incremental information; calibrate reliability flags; degrade Kepler observations to TESS-like duration/noise; compare with actual TESS sectors; quantify domain shift and possible recalibration.

Deliverables. D7.1 hybrid model and model card; D7.2 mission-transfer report; D7.3 third manuscript; D7.4 decision on broader TESS sample.

Exit criterion. A positive result requires locked, out-of-sample practical improvement or calibrated failure prediction. Otherwise, the publishable conclusion is a quantified redundancy/ transfer-limit result.

6.1.9. WP8 – Synthesis, release, and thesis

Aim. Consolidate the work into durable scientific products.

Tasks. Finalize code/data releases and DOIs; reconcile terminology and assumptions across papers; prepare thesis chapters and defense materials; archive negative results, configuration files, and complete provenance; respond to peer review.

Deliverables. D8.1 released package; D8.2 archival data/figure bundle; D8.3 thesis; D8.4 defense seminar and post-thesis roadmap.

6.2. Stage gates

Table 6.1.: Decision gates that control scope and protect the critical path.

Gate	Target	Pass condition	Contingency if not passed
G0	Month 3	Reproducible Rudy baseline and validated spectral normalization	Pause TDA; resolve observation-model and provenance errors
G1	Month 7	Simulations recover nominal false-alarm/coverage behavior	Restrict representations; revise uncertainty model
G2	Month 12	At least one topology statistic survives nulls and simple ablations	Publish rigorous negative/benchmark result; focus on QC and bounds
G3	Month 19	Frozen LEGACY validation completed without leakage	Reduce model complexity; report subgroup-specific validity
G4	Month 24	Topological échelle relation/estimator meets theory or utility threshold	Retain cylindrical geometry as visualization; do not claim a new seismic observable
G5	Month 28	Papers/code archived and thesis synthesis complete	Freeze extensions; reserve remaining time for thesis and peer review

7. Expected results and publication strategy

7.1. Expected scientific results

The most plausible outcome is not that topology replaces existing asteroseismic pipelines. It is that carefully chosen topological summaries provide stable morphology, quality assurance, and one or two complementary observables while conventional spectral methods retain superior point estimation in high-S/N Kepler data.

1. **A validated topology of spectral peaks.** PSD H_0 will recover prominence exactly and provide a stable, threshold-free hierarchy of candidate modes and background structures. Its main gain should be calibrated feature survival and automated quality diagnostics.
2. **A topological oscillation-envelope estimator.** Persistence-weighted peak locations should recover $\nu_{\max}$ with competitive robustness under moderate background misspecification, though not necessarily lower variance than optimized classical smoothing.
3. **A cylindrical échelle score.** $\mathcal{T}(\Delta)$ is expected to peak near $\Delta\nu$ when at least several orders are visible. The method should be especially informative when ridge continuity is clearer than individual peak identification.
4. **Geometric recovery of separation ratios.** On high-quality LEGACY stars, persistent ridge localization should yield r_{02} or related circular separations with calibrated uncertainty. The likely contribution is a global ridge estimator, not a replacement for detailed peak-bagging posteriors.
5. **A coherence/damping diagnostic with a strict null.** Delay H_1 may separate coherent/quasi-periodic contributions from phase-randomized signals with the same PSD. A positive result would add time-domain information unavailable to PSD-only global pipelines; a null result would show that the apparent topology is determined by second-order spectral structure.
6. **Operating-domain maps.** Simulations should determine the minimum duration, S/N, mode lifetime, duty cycle, and number of visible radial orders required for each statistic.
7. **Formal physical bounds.** The thesis should deliver exact idealized relations and useful perturbation inequalities such as equation (4.29), explicitly distinguishing them from empirical regressions.
8. **A measured complementarity result.** Hybrid models may improve reliability/failure prediction more strongly than raw parameter accuracy. This would still constitute a valuable contribution for automated survey-scale analysis.

7.2. Quantitative success criteria

Table 7.1.: Primary success criteria. Final numerical margins are frozen after WP2 pilots and before validation.

Outcome	**Minimum success criterion**	**Stronger outcome**
Rudy reproducibility	Baseline estimates fall within joint documented reference uncertainty; all intermediate products reproducible	Independent topological estimates agree and yield calibrated diagnostic advantages
$\nu_{\max}$ topology	Bias controlled in simulations; interval coverage near nominal; no unacceptable catastrophic-failure increase	Better robustness than smoothing under background/window perturbations
$\Delta\nu$ topology	Correct alias selected in a prespecified fraction of detectable simulations and LEGACY stars	Competitive error plus lower false-confidence rate in low-S/N regimes
Ridge ratios	Agreement with reference ratios within calibrated uncertainty for adequate-S/N subset	Useful ratios recovered without individual mode labels
Hybrid value	Positive out-of-sample change beyond a locked practical-equivalence margin	Improvement across accuracy, calibration, and transfer
Theory	One nontrivial exact identity/bound with explicit assumptions and verified numerical tests	Bound guides cadence/duration/S/N or preprocessing design
Reproducibility	Public, tested package; manifests/configurations; archived results and DOI	Independent reproduction by a collaborator or external group

7.3. Anticipated figures and tables

The papers and thesis will emphasize diagnostic, falsifiable visuals:

- Rudy light curve, window function, PSD/background, ordinary échelle, and topological échelle in a common frequency convention;
- persistence diagrams annotated by physical frequency location rather than anonymous points;
- $\mathcal{T}(\Delta)$ with half/double-spacing aliases and bootstrap distribution;
- simulated detection/coverage maps over duration, S/N, lifetime, and duty cycle;
- surrogate distributions demonstrating which statistics contain phase information;
- paired error and calibration comparisons on the LEGACY holdout;
- ridge-angle versus published r_{02} with posterior predictive intervals;
- Kepler-to-TESS degradation curves and actual-sector placement;
- a failure atlas showing examples, reason codes, and recommended interpretation.

7.4. Publication plan

Table 7.2.: Planned manuscripts. Titles are provisional.

Paper	Core claim	Minimum publishable content	Candidate venue
I	Persistent topology of solar-like oscillation spectra: the Rudy benchmark	Analytic peak relation; validated simulation/null tests; classical and topological Rudy analysis; open code	MNRAS, A&A, ApJ
II	A cylindrical topological échelle diagram and validation on Kepler LEGACY stars	Formal construction; spacing/ridge estimators; ablations; locked population validation; failure domains	A&A, MNRAS, ApJ
III	Complementary topology and cross-mission reliability for solar-like asteroseismology	Hybrid inference; Kepler-to-TESS degradation; actual TESS tests; calibrated quality flags	AJ, ApJS, MNRAS
IV (optional)	Stability bounds and persistent ridge geometry for noisy comb spectra	Proofs, counterexamples, uncertainty inequalities, periodic/zigzag algorithms	Foundations of Data Science or methodological astronomy venue

The journal decision will follow the final balance between astronomical discovery, reusable method, and catalog value. Paper I should be submitted early enough that referee feedback can improve the population analysis. Software and data papers are not counted as substitutes for the principal astronomy manuscripts.

7.5. Value of negative results

Several outcomes are scientifically publishable if established rigorously: persistence adds no information beyond prominence; delay topology is fully predicted by the PSD; the topological échelle score reduces to a simpler ridge statistic; or TESS baselines are insufficient for Rudy-like high-frequency oscillations at the observed noise. Such results would prevent overclaiming in an emerging intersection, provide benchmark datasets, and clarify where topology is or is not warranted.

8. Risks, limitations, and research integrity

8.1. Risk register

Table 8.1.: Technical and programmatic risks. Probability (P) and impact (I) are rated low (L), medium (M), or high (H).

Risk	P	I	Mitigation	Trigger / fallback
TDA is redundant with classical statistics	M	H	Mandatory topology-free ablations; diagram localization; incremental-information tests	Reframe around exact equivalence, stability, QC, and quantified redundancy
Rudy TESS data have inadequate S/N or duration	H	M	Treat TESS as transfer test; use sector combination cautiously; degrade Kepler	Publish an upper limit/detectability analysis; do not force a seismic result
Spectral-window topology mimics oscillations	M	H	Real-window simulations; background-only nulls; window-feature annotations	Restrict claims to regimes passing maximum-statistic null tests
Preprocessing dominates diagrams	M	H	Registered branch analysis; transfer-function tests; pipeline-uncertainty term	Use topology primarily as a preprocessing diagnostic
Persistence hyperparameters overfit 66 stars	M	H	Simulation-first tuning; low-dimensional summaries; nested grouped validation	Simplify to prespecified statistics; abandon complex learners
$\Delta\nu/2$ or $2\Delta\nu$ aliases dominate	M	M	Joint ridge-count/degree geometry; prior physical interval; explicit alias output	Report multimodal posterior/reliability rather than a forced point estimate
Mode lifetimes destroy delay loops	H	M	Derive finite-coherence expectations; windowed/zigzag analyses; PSD-preserving surrogates	Record a negative phase-information result
Theoretical bounds are too loose	M	M	Normalize physically; localize perturbations; derive model-specific/probabilistic bounds	Retain formal bounds as guarantees, emphasize empirical calibration

Risk	P	I	Mitigation	Trigger / fallback
MESA/GYRE scope expands excessively	M	M	Compact designed grid; reuse public grids where suitable; stage-gate models	Limit to structural stress tests around observed sample
Compute/storage bottleneck	L–M	M	Profile pilots; chunk diagrams; cache immutable intermediates; batch parallelism	Reduce factorial grid using adaptive design and emulator
Publication timeline exceeds doctorate	M	H	Early Paper I; manuscript-ready WP deliverables; freeze extensions at G4	Prioritize two core papers and thesis; defer TESS expansion/optional theory paper

8.2. Known scientific limitations

- Persistent homology is stable to some perturbations but not automatically invariant to scientifically important changes in normalization, detrending, or sampling.
- The usual scalar filtration compresses amplitude, frequency, and order information; localization and graph geometry are needed to recover them.
- A 66-star population supports validation of compact hypotheses, not unconstrained representation learning.
- Scaling relations carry astrophysical systematics independent of estimator precision, including reference-value and evolutionary corrections.
- TESS transfer is affected by duration, dilution, passband, cadence, systematics, and sector selection simultaneously; domain-shift attribution requires degradation experiments.
- Solar-like stochastic excitation means repeated realizations of the same star can have materially different peak topology. Inference must target distributions, not a fixed diagram.
- Early subgiants and red giants may require multiparameter or zigzag constructions because mixed modes violate the simple p-mode ridge model.

8.3. Research integrity and open-science plan

The data are public astronomical observations and simulations; no human participants, personal data, or animal research are involved. The relevant integrity risks are selective reporting, data leakage, hidden preprocessing flexibility, and irreproducible computation. The project will address them through:

1. preregistration of primary hypotheses, metrics, exclusion rules, and the final holdout procedure;
2. a complete analysis ledger, including failed representations and negative controls;
3. versioned data manifests and checksums instead of mutable local filenames;
4. clear separation of exploratory, tuning, and confirmation partitions;
5. public code under an OSI-approved license, archived releases with DOIs, and citeable machine-readable outputs;

6. figure scripts that regenerate every numerical panel from archived tables;
7. reporting of computational cost and environmental assumptions;
8. authorship and contribution statements agreed at manuscript initiation.

Any use of generative tools for code or prose will follow institutional and journal policies, be reviewed by the researcher, and never substitute for verification of mathematics, citations, or results.

8.4. Data-management plan

Raw mission data remain identifiable by MAST product IDs and checksums. Small manifests, configurations, derived tables, posterior summaries, and figure inputs will be preserved in a DOI-bearing repository. Large reproducible intermediates may be omitted if their regeneration command and parent checksums are archived. Proprietary data are outside the planned core study. Published catalogs will include units, definitions, null values, uncertainty semantics, and quality flags following FAIR principles.

9. Schedule and agenda

9.1. Calendar alignment and doctoral critical path

Defense constraint. The researcher is currently targeting a May 2027 doctoral defense. If M1 is August 2026, the executable doctoral window is ten months. The accelerated plan below therefore prioritizes a defensible core contribution: analytic foundations, simulation/null calibration, the Rudy benchmark, a compact locked LEGACY validation, Paper I, and the thesis. The 30-month Gantt that follows represents the full publication program; work after M10 is post-defense continuation unless the institutional schedule changes.

Table 9.1.: Accelerated doctoral critical path to a May 2027 defense.

Month	Calendar	Locked activity and evidence	Output
M1	Aug 2026	Protocol, literature map, repository, MAST manifests, exact benchmark definitions	D0 and preregistration
M2	Sep 2026	Reproduce Lightkurve/pySYD Rudy baseline; window/background audit; ordinary echelle	G0 baseline
M3	Oct 2026	Analytic sinusoid/peak cases; simulator and surrogate validation	Analytic test suite
M4	Nov 2026	PSD and delay topology; null calibration; topology-free ablations	Locked pilot statistics
M5	Dec 2026	Cylindrical echelle prototype; full Rudy analysis; uncertainty branches	Rudy result freeze
M6	Jan 2027	Process a compact development subset of LEGACY stars; repair pipeline failures	Population QC freeze
M7	Feb 2027	Open the prespecified compact validation set; paired performance and calibration	Confirmation tables
M8	Mar 2027	Write and submit Paper I; freeze thesis figures, proofs, and negative results	Manuscript submission
M9	Apr 2027	Complete thesis synthesis; independent reproduction; committee pre-defense review	Thesis submitted
M10	May 2027	Corrections, archival release, defense seminar, and contingency buffer	Defense and v1.0 release

Decision criterion

The accelerated thesis should not promise a fully optimized hybrid model, exhaustive MESA/GYRE grid, broad TESS population, or three completed papers before May 2027. Those remain scientifically valuable extensions. The core defense claim is narrower and stronger: a mathematically defined, null-calibrated topological asteroseismology framework demonstrated on Rudy and tested on an independently locked Kepler cohort.

9.2. Full thirty-month publication program

The full schedule is expressed relative to project month M1. Its critical path is baseline → simulation calibration → locked population validation → theory/hybrid synthesis → thesis and archival release. Paper writing overlaps analysis rather than being deferred to the end.

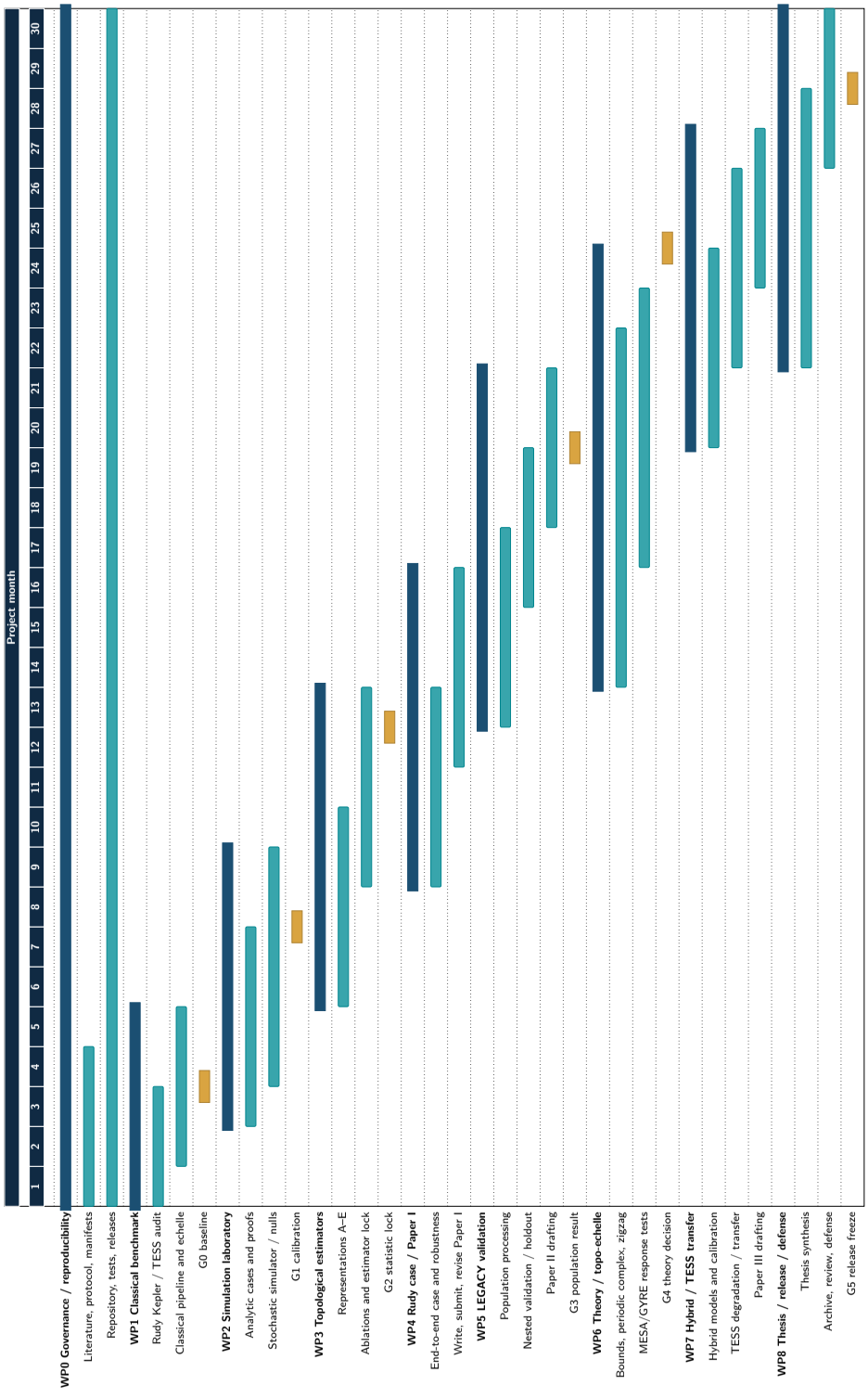


Figure 9.1.: Thirty-month Gantt plan. Continuous reproducibility work and overlapping writing reduce end-loaded risk.

9.3. Milestones and review agenda

Table 9.2.: Milestones, evidence, and review decisions.

Month	Milestone	Evidence reviewed	Decision
M1	Project launch	Protocol, roles, repository, resource access	Approve definitions and critical path
M3	Baseline gate G0	Reproduced Rudy parameters; data-quality audit; TESS inventory	Proceed to topology or repair observation model
M5	Simulation pilot	Analytic regression tests; simulator PSD/mode validation	Freeze core factors and nulls
M7	Calibration gate G1	Type-I error, coverage, persistence stability, runtime profile	Scale experiments or revise uncertainty
M9	Representation review	Redundancy matrix, interpretable localizations, initial ablations	Retain/drop representations
M12	Statistic gate G2	Locked estimators, null thresholds, Rudy blinded outputs	Authorize confirmation analysis
M14	Paper I submission	Complete reproducibility capsule and internal review	Submit; incorporate feedback into WP5
M16	LEGACY processing review	Exclusion ledger, QC distributions, reference reconciliation	Freeze population cohort
M19	Population gate G3	Nested validation and final holdout results	Set Paper II claims and scope
M21	Paper II submission	Theory/validation manuscript and released benchmark	Submit; freeze population expansion
M24	Theory gate G4	Proofs, counterexamples, topological echelle utility	Decide optional math paper versus thesis chapter
M26	TESS/hybrid review	Incremental-information and domain-shift results	Set Paper III conclusion
M27	Paper III submission	Cross-mission manuscript, code/model card	Submit and freeze new analysis

Month	Milestone	Evidence reviewed	Decision
M28	Release gate G5	Archived package, data tables, thesis near-final	Only corrections and defense work remain
M30	Thesis/defense	Final thesis, seminar, responses, long-term archive	Complete doctorate and post-thesis roadmap

9.4. Recurring working rhythm

- **Weekly:** code/data-quality review, experiment ledger update, and one reproducible result checkpoint.
- **Biweekly:** supervisory meeting focused on one decision, not an undifferentiated status report.
- **Monthly:** milestone dashboard covering scientific evidence, risks, compute, writing, and open-review actions.
- **Quarterly:** red-team review of leakage, null hypotheses, negative results, and scope; update the risk register.
- **Before every submission:** independent reproduction from a clean environment, citation audit, figure/table data audit, and consistency check between claims and registered criteria.

10. Conclusion

This project is ambitious but coherent because it treats topology as a mathematical measurement layer embedded within a mature physical discipline. Its strongest route to publishable quality is the combination of exact small results, realistic stochastic validation, an exceptionally transparent benchmark, a locked population test, and candid topology-free ablations. The cylindrical topological échelle diagram is the central high-risk/high-reward idea; the broader thesis does not depend on its success.

By the end of the doctorate, the project should answer four questions with evidence: what structures in a solar-like light curve or spectrum have stable topology; which standard seismic quantities admit exact or calibrated topological counterparts; whether those counterparts improve inference or reliability; and under which observing conditions the answer changes. A positive result establishes a new complementary tool for survey asteroseismology. A well-calibrated negative result establishes equally valuable boundaries for the responsible use of TDA in astronomy.

A. Notation and operational definitions

Table A.1.: Principal symbols and operational definitions.

Symbol	Name / unit	Operational meaning
$x(t)$	relative flux	Quality-masked, normalized observation with preprocessing branch recorded
$P(\nu)$	PSD	One-sided power-density estimate with stated normalization and window
$B(\nu)$	background PSD	Posterior/predictive Harvey-like background including white noise
$S(\nu)$	normalized spectrum	P/B or $\log P - \log B$, definition fixed per experiment
$\nu_{\max}$	μHz	Center of the solar-like oscillation power excess under declared estimator
$\Delta\nu$	μHz	Mean radial-mode spacing over the fitted frequency interval
$\delta\nu_{02}$	μHz	$\nu_{n,0} - \nu_{n-1,2}$ under the adopted order convention
r_{02}	dimensionless	$\delta\nu_{02}/\Delta\nu$ or published local-ratio convention, explicitly stated
Dgm_k	persistence diagram	Multiset of birth–death pairs for homology degree k
π_i	persistence	$d_i - b_i$ in filtration units
d_{B}	bottleneck distance	Diagram distance under optimal matching including diagonal
E_Δ	echelle map	Periodic map from frequency to order q and angle θ
$\mathcal{T}(\Delta)$	topological echelle score	Prespecified ridge-survival/concentration statistic with penalties
$\widehat{\nu}_{\max}^{\mathrm{top}}$	topological estimator	Persistence-weighted oscillation-envelope location
$\widehat{\Delta\nu}^{\mathrm{top}}$	topological estimator	Maximizer of a locked comb or echelle persistence score

B. Rudy benchmark checklist

☐ Resolve KIC/TIC identifiers and save complete MAST product manifest.

☐ Download Kepler short-cadence quarters 2, 5, 6, and 7 with checksums.

☐ Inventory TESS SPOC sector products; verify cadence and contamination metadata.

☐ Reproduce Lightkurve tutorial values and record package/reference constants.

☐ Fit independent background and global seismic parameters with pySYD.

☐ Plot and quantify spectral window and preprocessing transfer functions.

☐ Produce ordinary échelle diagram with blind trial-spacing diagnostic.

☐ Compute all five topological representations and localized diagrams.

☐ Run background-only, phase-randomized, IAAFT, and injection controls.

☐ Bootstrap/posterior-predict the complete topology selection pipeline.

☐ Compare topology-free ablations and document failures.

☐ Degrade Kepler data to TESS-like duration/cadence/noise before actual TESS comparison.

☐ Generate manuscript figures and tables from archived machine-readable outputs.

☐ Reproduce from a clean environment and archive the release.

C. Minimum simulation and ablation matrix

Table C.1.: Core experiments that cannot be omitted from the thesis.

Claim	**Varied factors**	**Null / ablation**	**Primary metric**
Topological $\nu_{\max}$	S/N, background slope, envelope width, duration	Smoothed-envelope estimator; background-only	bias, RMSE, coverage
Topological $\Delta\nu$	visible orders, curvature, lifetime, aliases, window	ACF, PS$\otimes$PS, contrast-only ridge score	alias accuracy, relative error
Delay coherence	lifetime, re-excitation, beating, gaps, cadence	phase randomization, IAAFT, PSD-only predictor	power at fixed false alarm
Ridge ratios	mode uncertainty, missing orders, rotation, curvature	circular mean/graph without persistence	bias, coverage, failure rate
Stability bound	bounded/adversarial and stochastic perturbations	empirical diagram change versus theoretical bound	tightness ratio
Hybrid value	star type, S/N, magnitude, mission	classical-only; topology-only; shuffled topology	paired log score, calibration
TESS transfer	duration, noise, dilution, sector gaps	matched Kepler degradation, actual sector window	domain-shift error

D. Reproducibility and reporting checklist

- ☐ Data IDs and checksums published
- ☐ Quality masks defined
- ☐ Units and PSD normalization stated
- ☐ Sampling window plotted
- ☐ Preprocessing transfer tested
- ☐ Random seeds recorded
- ☐ Dependency lock archived
- ☐ Topology metric/filtration defined
- ☐ Coefficient field declared
- ☐ Hyperparameters selected without leakage
- ☐ Star-grouped validation used
- ☐ Complete-selection bootstrap used
- ☐ Null/surrogate generation documented
- ☐ Multiple testing addressed
- ☐ Effect sizes and intervals reported
- ☐ Negative results retained
- ☐ Failure cases cataloged
- ☐ Classical baselines reproduced
- ☐ Topology-free ablations included
- ☐ Physical interpretation localized
- ☐ Scaling covariance propagated
- ☐ Holdout opened once
- ☐ Figure inputs machine-readable
- ☐ Clean-environment reproduction passed
- ☐ Code/data DOI provided
- ☐ Claims matched to success criteria

E. Future extensions outside the critical path

If the core objectives are completed early, the framework can be extended to mixed modes and red giants through zigzag or multiparameter persistence; activity-cycle changes through vineyards; population morphology through diagram barycenters; and joint photometry/spectroscopy through multiview topology. Other promising directions include persistence-based acoustic-glitch detection, uncertainty-aware fibered barcodes for simultaneous amplitude/frequency thresholds, and differentiable topological losses for physically constrained ridge extraction. These topics are explicitly deferred so they do not displace the validation work required for credible astronomy publication.

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